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Thickness dependent structural, magnetic, and electronic properties of the epitaxial films of transparent conducting oxide $NiCo_2O_4$

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We present the thickness dependent structural, magnetic, and transport properties of transparent conducting spinel $NiCo_2O_4$ thin films on $MgAl_2O_4$ (001) substrates. The structural examination of the films reveals that the epitaxial stain is independent of the films' thickness. Electric and magnetic measurements show that the films are metallic with p-type conduction and ferrimagnetic down to 2 unit cells with an enhanced coercive field in the films thinner than 30 unit cells. The low-temperature resistivity data indicate that the observed resistivity minimum results from the disorder-induced quantum interference effects. Our results demonstrate that $NiCo_2O_4$ may provide an alternative magnetic conducting medium for spintronics devices. © *2013 AIP Publishing LLC*. [http://dx.doi.org/10.1063/1.4820930]

I. INTRODUCTION

Magnetic spinel films, such as $NiFe_2O_4$ and $CoFe_2O_4$ ferrites, have been widely investigated due to their robust magnetic properties with the potential for spin injection and/or filtering applications in spintronics.[1–8] The attraction of the ferrites is that they retain and even enhance[1,6,8] their magnetism in ultrathin few-unit-cell films; their disadvantage is that they are semiconductors and need to be doped[8] to conduct electric current. The spinel $NiCo_2O_4$ has long been known to possess ferrimagnetic order and the possibility of band-like p-type conduction has been suggested for the material.[9–11] Here, we show that ultrathin epitaxial $NiCo_2O_4$ films remain magnetic and metallic down to the thickness of 3 unit cells (uc). These newly observed excellent properties may open the door for the use of $NiCo_2O_4$ in spintronics devices as storage or a spin-injection/filtering medium.

Historically, the spinel $NiCo_2O_4$ has been studied for its remarkable optical, electrical, and catalytic properties for various technological purposes including fuel cell electrodes, oxygen catalysis, reduction in alkaline media, water electrolysis, and for sensors and flat panel displays as a transparent conducting oxide.[11–14] The material is prepared most often in the form of polycrystalline particulate samples with a rather low packing density or in the form of polycrystalline thin films,[15] which makes it difficult to characterize its intrinsic properties.[11,15] In our previous study, we reported the synthesis of high-quality epitaxial $NiCo_2O_4$ films using pulsed laser deposition (PLD) method. The films grown at temperatures of 300–400 °C show metallic behavior accompanied with ferrimagnetic order with the Curie temperature $T_c \sim 300$ K, while increasing the growth temperature above 500 °C tunes the conductivity of the films from metallic to insulating and renders the films non-magnetic.[16] The magnetic properties of $NiCo_2O_4$ have not been used in applications but have been investigated in the context of the mixed valencies of the Ni and Co cations in this inverse spinel, where the Ni cations occupy almost exclusively the octahedral sites and the Co cations are distributed evenly between the tetrahedral and octahedral sites.[9,10] Various models have been proposed that put the cation oxidation states in the (2.5–3)+ and (2.5–3.5)+ range for the octahedral Co and Ni, respectively.[9,10] In applied magnetic field, the mixed-valence cationic character leads to a redistribution of charge and a modification of the oxidation states and of the spontaneous magnetic moment. In our $NiCo_2O_4$ films grown at 350 °C, we reported a field-dependent spontaneous magnetization of 1.87 μ_B per formula unit in the field 2 T, consistent with previous reports.[9,10] The transport and magnetic properties of thin films depend strongly on their thickness at the ultrathin level. As the thickness becomes smaller, such observables as the resistivity, saturation magnetization, and coercive field can be influenced to a high degree by the films' geometrical boundaries, interfacial scattering, and pinning, and the crossover from three-dimensional to two-dimensional physics.[17–19] A thorough understanding of a material's behavior as the function of film thickness is indispensable before it can be integrated in technological applications. We have investigated the evolution of the resistivity and the magnetic properties of $NiCo_2O_4$ in films down to 1 uc in thickness and found enhanced coercivity in films below 30 uc. The films of 2 uc and less become highly resistive and lose the long-range magnetic order, and we regard the 2 uc as the thickness of the electric and magnetic dead layers. All films thicker than the dead layer displayed metallic behavior below room temperature with a low-temperature resistivity minimum at $T_{min} \sim 50$–60 K. The analysis of the temperature dependence of the resistivity below T_{min} points to the

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dominant role of quantum interference effects (QIE) in the transport in $NiCo_2O_4$ at low temperatures and in the ultrathin regime.

II. EXPERIMENTAL DETAILS

$NiCo_2O_4$ films of thickness ranging from 1 uc to 228 uc (1 uc = 0.8114 nm) were grown epitaxially on $MgAl_2O_4$ (001) substrates by PLD at the constant substrate temperature of 350 °C and the 50 milli Torr O_2 pressure inside the chamber. The structural properties were analyzed using four circle high resolution X-ray diffraction (HR-XRD), while the thicknesses of the films were determined by the x-ray reflectivity (XRR) measurement. The magnetic properties of the films were measured using a superconducting quantum interference device (SQUID) magnetometer. The resistivity was measured using the standard 4-probe method and the Hall resistivity was measured with magnetic field applied perpendicular to the film surface. Using the conventional photolithography and Ar ion milling, appropriate current and voltage leads were patterned on the films for the measurements of longitudinal and transverse resistivity.

III. RESULTS AND DISCUSSION

Figure 1(a) shows XRD 2θ-θ scans through the (004) symmetric reflections of the film and substrate for the 11, 30, 77, and 224 uc thick films, respectively. The intensity of the film peaks increases significantly as the thickness increases, but the peak position remains at the same value of 2θ. Figures 1(b)–1(d) show the reciprocal space mapping (RSM) scans around the (226) reflection of the films with thicknesses of 11, 225, and 875 uc. The horizontal and vertical peak positions for all the films are the same, indicating identical lattice parameters and the strain state independent of the films' thickness. The vertical position of the film peak is also identical to that of the substrate, showing that the films are fully strained up to the highest thickness (875 uc, 700 nm). The in- and out-of-plane lattice constants calculated from the x and y positions of the film peak are 8.076 Å and 8.173 Å, respectively. The smaller in-plane and the larger out-of-plane lattice constant value than the bulk indicates the compressive in-plane strain.

Figure 2(a) shows the temperature dependent resistivity, ρ(T), for the films with thicknesses 3, 11, 30, 77, and 224 uc. Even though we find a slow increase in resistivity as the film thickness decreases from 224 to 3 uc, the films maintain their metallic behavior below room temperature from 224 uc to 3 uc, with the lowest room temperature resistivity of ~1.4 mΩ cm for the 224 uc thick films. The observation of the metallic resistivity in the 3 uc film is possible due to its high crystalline quality. The room temperature Hall Effect measurements (not shown here) revealed p-type conduction. A region with dominant anomalous Hall effect was found in low magnetic fields (below 0.8 kOe).

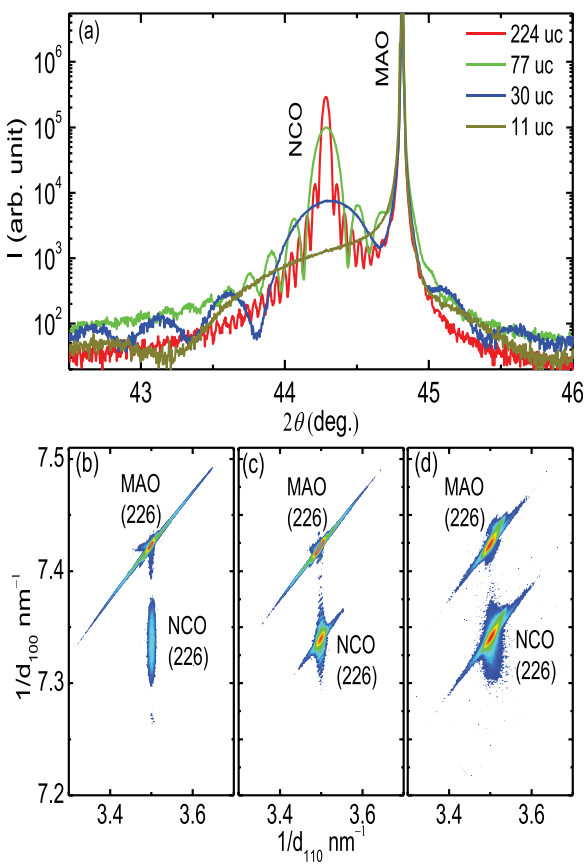

FIG. 1. (a) HR-XRD 2θ-θ scans of $NiCo_2O_4$ films on $MgAl_2O_4$ (001) substrate with the thicknesses of 11, 30, 77, and 224 unit cells. (b), (c), and (d) HR-XRD reciprocal space maps around the (226) reflections of the $NiCo_2O_4$ films on $MgAl_2O_4$ substrate with the thickness 11, 224, and 875 unit cells, respectively.

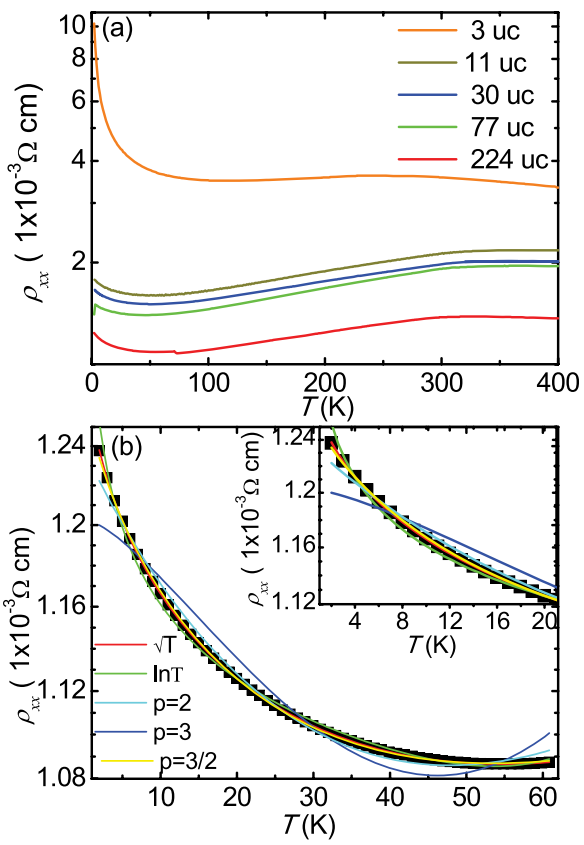

FIG. 2. (a) In-plane resistivity ρ vs. temperature T for the $NiCo_2O_4$ with the thicknesses of 3, 11, 30, 77, and 224 unit cells. (b) Resistivity ρ(T) for the 224 uc film and the fitted curves for Eqs. (1)–(3). The inset shows the magnified scale.

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The insight in the nature of electronic conduction in $NiCo_2O_4$ at low temperatures and few-unit-cell thicknesses can be gained by examining the temperature dependence of resistivity below T_{min} (Fig. 2). Similar-looking resistivity minima are found in a wide range of magnetic materials and have been attributed to the Kondo effect,[20] magnetic cluster scattering,[21] spin-polarized tunneling through grain boundaries,[22] and QIE via the Coulomb interaction between electrons enhanced by disorder or the weak localization.[23–26] Spin-polarized tunneling through grain boundaries is significant only in polycrystalline samples and we exclude it from consideration here. To consider the possibility of Kondo-like spin-dependent scattering as the source of the resistivity minimum in our $NiCo_2O_4$ films, we fit the temperature dependence of ρ(T) in the 2–60 K range with the functional form

$$\rho(T) = \frac{1}{\sigma_0 + C\ln T} + AT^n, \tag{1}$$

where all scattering processes, such as electron-phonon, electron-magnon, and electron-electron, are described by the single power law ($\mathbf{A}T^n$) term, and σ_0, C, A, and n are fitting parameters. The fitting was carried out for all metallic films and as an illustration, the best fit for the 224 uc film is shown in Fig. 2(b). As discussed below, the logarithmic temperature dependence of the low-temperature resistivity in Eq. (1) does not provide the best fit of the data. Therefore, we do not consider Kondo-like scattering as the cause of the resistivity upturn below T_{min}. Such conclusion is consistent with our earlier observation that the low-temperature upturn is more pronounced in high-temperature-grown non-magnetic $NiCo_2O_4$ films,[16] and also speaks against the magnetic cluster scattering as the dominant physics behind the minimum resistivity. The logarithmic temperature dependence of Eq. (1) could also describe the residual conductivity governed by the disorder-enhanced electron-electron (e-e) interactions or the weak localization in the 2D limit. The unsatisfactory fit to Eq. (1) allows us to exclude such 2D QIE as well.

$NiCo_2O_4$ is an intrinsically disordered material, as Ni and Co cations are randomly distributed over the occupied octahedral sites. The disorder-induced localization tendencies in the 3D limit are described by the following equations:

$$\rho(T) = \frac{1}{\sigma_0 + B\sqrt{T}} + AT^n, \tag{2}$$

$$\rho(T) = \frac{1}{\sigma_0 + BT^{p/2}} + AT^n, \tag{3}$$

where σ_0, B, A, and n are fitting parameters. Equation (2) describes the localization due to disorder-enhanced e-e interaction, while Eq. (3) describes the weak localization with the parameter p dependent on the dominant scattering mechanism:[25–28] $p = 3/2$ for the e-e scattering in the dirty limit, $p = 2$ for the e-e scattering in the clean limit, and $p = 3$ for the electron-phonon scattering. As Fig. 2(b) shows, the best fits of the measured data are achieved with Eq. (2) and Eq. (3) with $p = 3/2$. The fits using Eq. (3) with p = 2 and p = 3 are not adequate. Therefore, we propose that disorder-induced QIE in the 3D limit are responsible for the resistivity minimum in $NiCo_2O_4$, although the present data do not allow us to distinguish between the contributions of the Coulomb interaction and the weak localization. The Coulomb interaction should result in a positive magnetoresistance, while we expect negative magnetoresistance from the weak localization. In the $NiCo_2O_4$ films, we reported negative magnetoresistance of 2% in the field of 1 T.[16] However, this magnetoresistance also includes the effects of the double-exchange, which was proposed as the mechanism of the metal-insulator transition in $NiCo_2O_4$ (Ref. 11) and which may mask the contributions of the QIE to the magnetoresistance. Since the two contributions to the QIE generally produce effects of the same magnitude, we believe that they both are present in the studied $NiCo_2O_4$ films, even though the weak localization is likely to become stronger in the ultrathin films.[29]

The magnetic measurements on the $NiCo_2O_4$ films of various thicknesses are summarized in Figs. 3(a) and 3(b). We have subtracted the diamagnetic contribution from the substrate by a separate measurement. Figure 3(a) shows the magnetization m(T) as a function of temperature measured in the 2 kOe field for the 3, 11, 30, 77, and 224 uc films with no significant change in the Curie temperature T_c with decreasing thickness[30] down to 11 uc. Figure 3(b) displays the m(H) hysteresis curves of these films measured from at 10 K. For the films thinner than 30 uc, H_c increases significantly (the inset of Fig. 3(a)) and reaches the maximum of ~1500 Oe in the 3 uc film. The coercivity is likely promoted by the decreasing film thickness as the relative strength of the surface and/or interfacial domain wall pinning increases compared to the bulk magnetic energy. The dependence of the saturation magnetization on the film thickness is shown in the inset of Fig. 3(d); the magnetization begins to degrade below the thickness of 7 uc. The sheet conductance ($G\square$) at T_{min} shows a linear dependence with thickness with a finite

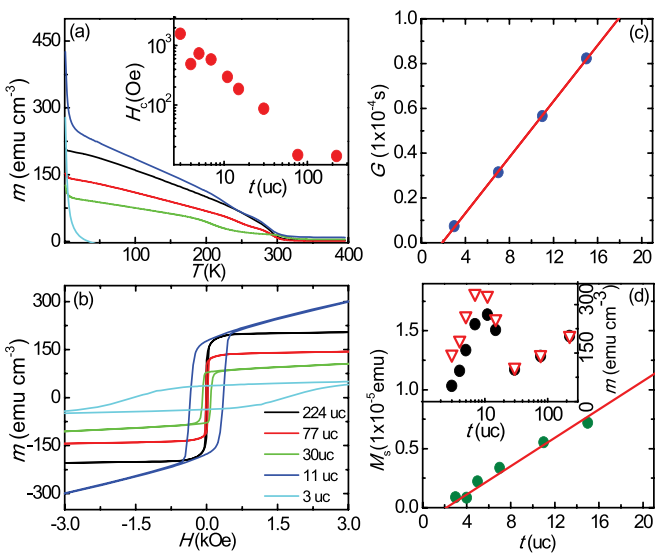

FIG. 3. (a) Magnetization m vs. temperature T for the $NiCo_2O_4$ films with the thicknesses of 3, 11, 30, 77, and 224 unit cells. The inset is the coercive field H_c vs. thickness t. (b) m vs. magnetic field H for the same set of films. (c) Sheet conductance $G\square$ vs. thickness t. (d) Saturation magnetic moment (M_s) at 5 kOe vs. thickness t. The inset shows saturation magnetization m at 5kOe vs. thickness t (closed circles) and ($mt/(t\text{-}t_{dead})$) vs. thickness t (open triangles). $t_{dead} = 2$ uc is the thickness of the magnetic dead layer.

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intercept of the x axis at 2 uc (Fig. 3(c)). Similarly, the magnetic moment $M_s = mV$ (V–film volume) decreases linearly with thickness and intersects the x axis at ~2 uc (Fig. 3(d)). Based on the two intercepts, we estimate the thickness of the electrical and magnetic dead layers to be 2 uc (1.62 nm). For comparison, the electrical dead layer thickness in ultrathin $La_{0.7}Sr_{0.3}MnO_3$ films was reported as 3.2 nm and 4 nm and the magnetic dead layer thickness as 1.2 nm and 10 nm.[31,32]

IV. SUMMARY

In summary, we presented the thickness-dependent structural, electronic, and magnetic properties of epitaxial $NiCo_2O_4$ films. The structural analysis revealed that the epitaxial strain in the $NiCo_2O_4/MgAl_2O_4$ (001) films is independent of the films' thickness up to 700 nm. We found that the films remain metallic and magnetic down to 2 uc in thickness. The resistivity minimum with $T_{min} \sim 50$–60 K is explained as the consequence of the disorder-induced quantum interference effects. We also observed the enhanced coercivity in thinner films, which could plausibly result from the domain wall pinning at the surface/interface. The enhanced magnetic properties and the metallicity make the ultrathin epitaxial $NiCo_2O_4$ films a good candidate material for spintronics, e.g., as a spin injection or magnetic tunnel junction electrode material.

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[1]S. Venzke, R. vanDover, and J. Phillips, J. Mater. Res. **11**, 1187 (1996).
[2]Q.-C. Sun, H. Sims, and D. Mazumdar, Phys. Rev. B **86**, 205106 (2012).
[3]M. Foerster, F. Rigato, K. Bouzehouane, and J. Fontcuberta, J. Phys. D: Appl. Phys. **43**, 295001 (2010).
[4]U. Luders, M. Bibes, and K. Bouzehouane, Appl. Phys Lett. **88**, 082505 (2006).
[5]A. V. Ramos, M.-J. Guittet, and J.-B. Moussy, Appl. Phys. Lett. **91**, 122107 (2007).
[6]U. Luders, M. Bibes, J. Bobo, M. Cantoni, R. Bertacco, and J. Fontcuberta, Phys. Rev. B **71**, 134419 (2005).
[7]J. A. Moyer, D. P. Kumah, C. A. F. Vaz, D. A. Arena, and V. E. Henrich, Appl. Phys. Lett. **101**, 021907 (2012).
[8]J. A. Moyer, C. A. F. Vaz, D. P. Kumah, D. A. Arena, and V. E. Henrich, Phys. Rev. B **86**, 174404 (2012).
[9]P. D. Battle, A. K. Cheetham, and J. B. Goodenough, Mater. Res. Bull. **14**, 1013 (1979).
[10]J. F. Marco, J. R. Gancedo, M. Gracia, J. L. Gautier, E. I. Rios, H. M. Palmer, C. Greaves, and F. J. Berry, J. Mater. Chem. **11**, 3087 (2001).
[11]P. Silwal, C. La-o-vorakiat, E. E. M. Chia, D. H. Kim, and D. Talbayev, "Effect of growth temperature on the terahertz-frequency conductivity of the epitaxial transparent conducting spinel NiCo2O4 films," AIP Advances (to be published).
[12]N. Bahlawane, P. A. Premkumar, J. Feldmann, and K. Kohse-Hoinghaus, Chem. Vap. Deposition **13**, 118 (2007).
[13]C. F. Windisch, G. J. Exarhos, K. F. Ferris, M. H. Engelhard, and D. C. Stewart, Thin Solid Films **398–399**, 45 (2001).
[14]M. N. Iliev, P. Silwal, B. Loukya, R. Datta, D. H. Kim, N. D. Todorov, N. Pachauri, and A. Gupta, J. Appl. Phys. **114**, 033514 (2013).
[15]J.-G. Kim, D. Pugmire, D. Battaglia, and M. Langell, Appl. Surf. Sci. **165**, 70 (2000).
[16]P. Silwal, L. Miao, I. Stern, X. Zhou, J. Hu, and D. H. Kim, Appl. Phys. Lett. **100**, 032102 (2012).
[17]R. K. Das, R. Misra, S. Tongay, R. Rairigh, and A. F. Hebard, J. Magn. Magn. Mater. **322**, 2618 (2010).
[18]V. D. Das and D. Karunakaran, Phys. Rev. B **30**, 2036 (1984).
[19]D. Samal, D. Venkateswarlu, and P. S. Anil Kumar, Solid State Commun. **150**, 576 (2010).
[20]P. Sarachik, E. Corenzwit, and L. D. Longinotti, Phys. Rev. **135**, A1041–A1045 (1964).
[21]T. C. Kuo and K. L. Wang, Appl. Phys. Lett. **59**, 3399 (1991).
[22]E. Rozenberg and M. Auslender, J. Phys. Condens. Matter **14**, 8755 (2002).
[23]A. Tiwari and K. P. Rajeev, Solid State Commun. **111**, 33 (1999).
[24]R. R. Jia, J. C. Zhang, R. K. Zheng, D. M. Deng, H.-U. Habermeier, H. L. W. Chan, H. S. Luo, and S. X. Cao, Phys. Rev. B **82**, 104418 (2010).
[25]P. Lee and T. Ramakrishnan, Rev. Mod. Phys. **57**, 287 (1985).
[26]D. Kumar, J. Sankar, J. Narayan, R. K. Singh, and A. K. Majumdar, Phys. Rev. B **65**, 094407 (2002).
[27]M. Ziese, H. C. Semmelhack, K. H. Hans, S. P. Sena, and H. J. Blythe, J. Appl. Phys. **91**, 9930 (2002).
[28]L. Maritato, C. Adamo, C. Barone, G. M. De. Luca, A. Galdi, P. Orgiani, and A. Yu. Petrov, Phys. Rev. B **73**, 094456 (2006).
[29]D. H. Kim, D.-W. Kim, B. S. Kang, T. W. Noh, D. R. Lee, K.-B. Lee, and S. J. Lee, Solid State Commun. **114**, 473 (2000).
[30]R. Zhang and R. Willis, Phys. Rev. Lett. **86**, 2665 (2001).
[31]R. P. Borges, W. Guichard, J. G. Lunney, J. M. D. Coey, and F. Ott, J. Appl. Phys. **89**, 3868 (2001).
[32]M. Huijben, L. W. Martin, and Y.-H. Chu, Phys. Rev. B **78**, 094413 (2008).

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